

SEI Report

A Decade of GHOSTS in the Machine: From Cyber Range Realism to Cognitive Mission Rehearsal

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Dustin D. Updyke
John W. Yarger
Thomas G. Podnar

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Abstract

GHOSTS began in 2017 as a practical response to a persistent cyber range problem: realistic cyber exercises require networks to be filled with traffic from users and supporting infrastructure. This provides data for exercise participants to analyze, but also provides a way for the adversary to blend in and hide its activity. Over the past decade, the project evolved from a host-based activity generator into a broader platform for simulating human, organizational, and machine behavior across cyber, social, and cognitive layers.

The central design principle of GHOSTS is *sufficient realism*. The system does not attempt to reproduce human behavior perfectly. Instead, it creates believable activity at the right fidelity to support training, experimentation, assessment, and mission rehearsal. This paper traces the evolution of GHOSTS as it grew from producing enclave activities (and the resultant network traffic, logs, endpoint artifacts, and browser activity) to creating non-player characters (NPCs) with preferences, relationships, knowledge, motivations, beliefs, workflows, content environments, and LLM-assisted decision making. It also describes the engineering lessons learned from sustaining GHOSTS as open source research software, including modularity, dependency management, documentation, observability, defaults, maintainability, and shared ownership.

As cyber operations increasingly intersect with broader operations in the information environment, GHOSTS now supports the larger mission of helping teams rehearse ideologically contested scenarios.

1 Introduction

Over the past decade, developing GHOSTS¹ has meant continually redefining the framework while improving it, driven by an evolving understanding of what sufficient realism requires.

GHOSTS research began in 2017 as a practical response to operational challenges in executing an almost year-long cyber training exercise series. We needed better ways to create the traffic, logs, and host artifacts that make a simulated environment feel alive. The participants in our exercises were trained blue team defenders, operators, and analysts that were smart enough to triangulate across many different signals, including endpoint behavior, application traces, user activity, timing, files, browser history, authentication events, and the audit trail real people leave behind.²

By this time, user simulation was not new, and the SEI had a great deal of experience in this area to draw upon, but the available tools were dated and often created problems. Some had performance issues during our long-running events, while others required installing non-standard software, adding dependencies, writing scripts, or making fragile host-level configuration changes. In defensive cyber operations environments, these were immediate red flags. Savvy exercise participants identified traffic patterns and leveraged them as filters for rapidly tracking the adversary. We noted that the tooling meant to create realism could itself become unrealistic, operationally fragile, or difficult to explain to the teams using the range.

The original effort behind GHOSTS emerged from that gap [Updyke 2018]. The earliest requirements were simple: build a proper compiled binary, make it remote controllable, and give range administrators a way to orchestrate NPCs,³ collect reports, and manage activity from a central server. This way, GHOSTS agents could run locally on hosts and perform their own activity, while a central API gave range administrators a single place to configure, assign, observe, and control what those agents were doing.

¹ <https://github.com/cmu-sei/GHOSTS>

² Blue teams defend the environment, red teams emulate adversaries attempting to compromise it, and purple teams integrate offensive and defensive perspectives to improve detection, response, and scenario learning.

³ A GHOSTS NPC is a configurable synthetic actor that operates within a simulated environment, carrying out realistic digital, social, or cognitive behaviors according to its assigned role, permissions, preferences, relationships, and beliefs.

This web-oriented architecture turned out to be one of the project’s strengths. The API, the data model, and the reporting mechanisms made it possible to orchestrate synthetic activity across many machines in a way that was scriptable, observable, and repeatable. The goal was not just to create traffic, but to make the environment produce enough believable evidence that participants could treat it as a living world.

Over time, we came to describe this goal as *sufficient realism*. We knew we could never perfectly reproduce human behavior inside a cyber range. Instead, we focused on creating enough believable activity, at the right fidelity, for the purpose at hand, whether that’s a mission rehearsal, an insider threat scenario, or a hunt exercise. Different use cases require different kinds of realism, so we do not attempt to simulate everything, just the specific factors participants need to observe, decide, and act upon in any given scenario.

The original problem we intended to address with GHOSTS persists—cyber ranges still need networks inhabited with routine activity and artifacts produced by plausible users. But the scope has expanded such that modern exercises must also represent the human and institutional systems through which technical events acquire meaning, spread, and affect decisions.

Table 1: *Expanding the Scope of Cyber Range Realism*

Layer	What the environment must represent	Examples
Technical environment	Ordinary activity across realistic, inhabited networks	Browser histories, file activity, host artifacts, application traces, authentication events, background traffic
Human and institutional environment	The factors that shape interpretation, behavior, and organizational response	Narratives, trust, public perception, decision confidence, institutional relationships, information flows
Adversary activity	Coordinated methods used to influence beliefs and behavior	Propaganda, disinformation, cyber activity, social media manipulation, troll farms, bot networks, state media, influence operations
Exploitable features of open societies	Democratic norms that adversaries treat as terrain for influence	Constitutional freedoms, rule of law, open dialogue, public debate

Adversaries operate across these layers simultaneously. They use technical systems, media channels, and social dynamics to shape what people believe, how they behave, and how organizations respond. In open societies, even foundational democratic values become surfaces that adversaries seek to exploit.

That broader framing is also visible in the evolution of joint and Department of War thinking about operations in the information environment (OIE) [Mulgund 2020, Theohary 2023, Maddox 2025]. As shown in Table 2, policy and doctrine have increasingly treated information as an operational concern tied to human perception, decision making, and behavior.

Table 2: *Evolution of Thinking on the Information Environment*

Year	Publication or Policy Action	Significance
2018	<i>Joint Concept for Operating in the Information Environment</i> [Joint Chiefs of Staff 2018]	Established a joint conceptual framework for operating in a contested information environment
2021	National Defense Authorization Act for Fiscal Year 2021 [U.S. Congress 2021]	Elevated the information environment as a matter of congressional and defense-policy concern
2022	<i>Joint Publication 3-04: Information in Joint Operations</i> ⁴	Formalized information as a joint operational concern involving effects on perceptions, attitudes, decisions, and behavior
2022	Government Accountability Office report on the information environment [GAO 2022]	Identified persistent challenges in organizing and coordinating information-related capabilities
2023	Government Accountability Office report on the contested information environment [GAO 2023]	Extended the analysis to the broader challenge of operating amid contested narratives, influence, and adversarial information activity

As information has changed from a supporting communications function to a core operational concern, the emphasis has shifted from message delivery toward effects on perceptions, attitudes, decisions, and behavior.

Given these changes in doctrine, GHOSTS has evolved to remain mission relevant. The practical implication is that cyber is no longer a matter of simply defending networks. A modern scenario such as the one depicted in Figure 1 may require a team of teams to defend networks and narratives together. A coordinated cyberattack on critical infrastructure, for example, is not only a malware problem. It may include manipulation of the domain name system (DNS), routing disruption, effects on supervisory control and data acquisition (SCADA), emergency service outages, hospital downtime, spoofed public alerts, rumors, panic, adversary messaging, and counter-messaging intended to preserve public trust. The network,

⁴ [Joint Chiefs of Staff 2022]

Joint Chiefs of Staff. *Joint Publication 3-04: Information in Joint Operations*. Joint Chiefs of Staff. 2022. Not publicly available. Accessed through Joint Electronic Library Plus. <https://jdeis.js.mil/>

the public reaction, the organizational response, and the information environment become one connected exercise space.

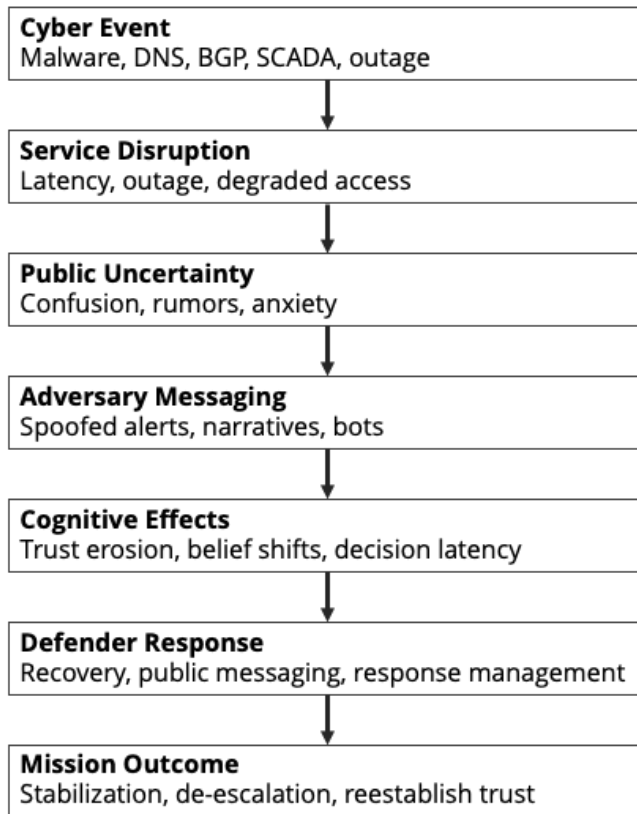


Figure 1: Example of a Multi-Stage Cyber-Social-Cognitive Mission

This paper reflects on that trajectory over a decade of work, over which GHOSTS grew from a way to generate practical traffic, logs, and artifacts to a broader platform for human, organizational, and machine activity simulation. It also highlights the engineering decisions that allowed GHOSTS to stay relevant long enough to become a widely used open-source software (OSS): dependency choices, modularity, documentation, and the slow accumulation of small improvements over years.

2 The Evolution of GHOSTS

When GHOSTS was first created, many other simulation tools in the training and exercise arena already existed, and the SEI had done a good amount of work directly in that domain. Many of the tools of that early era focused on traffic generation. Our first design decision in GHOSTS was to start not with traffic, but with synthetic activity created through the same kinds of applications, hosts, and artifacts that participants would expect to see in a real environment. If a user should generate network activity, that activity should come from an actual command or application.⁵ If a browser should be opened, GHOSTS should open a browser. If a document should be handled, GHOSTS should touch a real document.

In this way, GHOSTS began to generate the traffic, logs, endpoint behavior, and host artifacts of a believable environment where participants could do meaningful work. That goal shaped the architecture from the beginning. GHOSTS had to run on real hosts, drive real applications,⁶ and produce observable evidence that could survive scrutiny across packets, logs, files, browser history, authentication events, and other traces of user activity.

Table 3: *Capability Maturation Across GHOSTS Eras*

Year	Era	Main Capability Shift	New Realism Boundary to Consider
2017	GHOSTS	Host-based activity generation; Agent/NPC separation	Empty ranges needed believable background users; execution needed to be separated from representation.
2020	SPECTRE	Behavioral fidelity	Random browsing looked wrong.
2021	ANIMATOR	Synthetic people	Activity lacked a persona identity.
2021	Versions 6/7	Enterprise surface expansion	Adjacent workflows looked empty.
2022	Social, reasoning	Interaction and influence; Pandora content server	Posts and reactions lacked context; traffic lacked content diversity
2023	Early LLM work	AI-enabled action selection	Manually authored timelines did not scale.
2024	Version 8	Platform consolidation	Modularity created install and support friction.
2026	Version 9	Cyber-social-cognitive simulation	Endpoint realism alone was insufficient.

⁵ If a human user performing the same task would look identical when observed over the shoulder, the simulation was sufficiently realistic.

⁶ It should be noted that some applications actively avoid being automated for security purposes, making our simple goal sometimes difficult.

The broader mission has expanded, but the premise remains the same: the system must create activity that survives scrutiny. In early cyber range work, that evidence was mostly technical—packets, processes, files, browser activity, application traces, authentication events, downloads, and logs. In current cyber-social-cognitive work, the evidence also includes posts, messages, relationships, public reactions, belief changes, trust erosion, counter-messaging, and the timeline of how people and organizations respond to events.

2.1 Cyber Range Realism to Mission Rehearsal

A cyber exercise range is a controlled environment where networks, systems, applications, users, defenders, and adversaries interact in ways intended to emulate real operational conditions. Because many of these ranges are isolated or air-gapped, realism cannot depend on the live internet.⁷ The range must contain enough of the world to achieve exercise objectives.

Creating a believable cyber range requires replicating many layers at once, including network infrastructure, protocols and services, benign and malicious network traffic, middleware, applications, web content, user behavior, and a realistic threat landscape. Routers, switches, sensors, servers, DNS, HTTP, SMTP, file sharing, web browsing, email, document repositories, malware, attack vectors, and ordinary user behavior all become part of the synthetic environment. We describe this layered view of cyber range realism in detail in the technical report, *Foundation of Cyber Ranges*, emphasizing that realistic ranges must integrate infrastructure, services, traffic, applications, users, and threats into a coherent exercise environment [Podnar 2021].

⁷ The creation of an air-gapped internet involves networking, systems, and content creation highlighted in *Foundation of Cyber Ranges* [Podnar 2021].

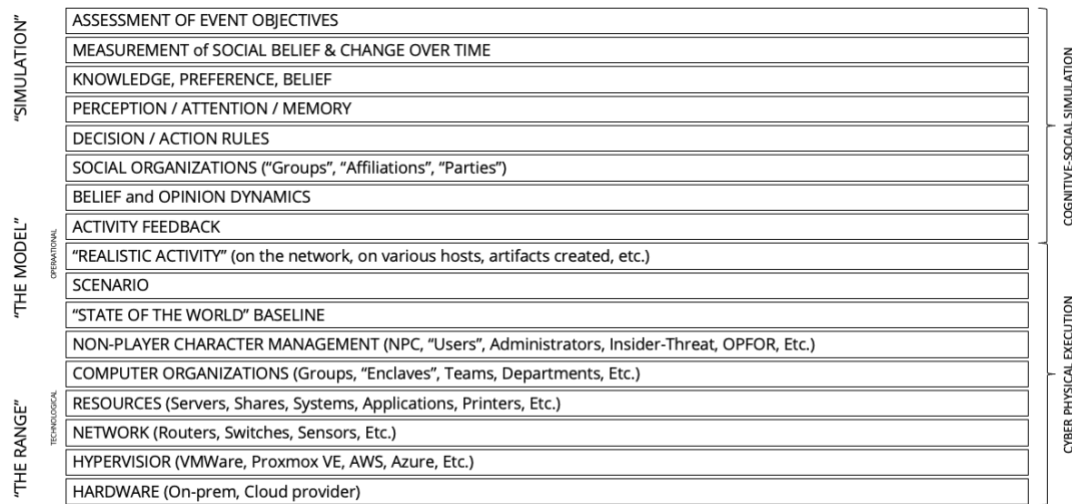


Figure 2: Cyber-Social-Cognitive Range Architecture

In Figure 2, as we move up the stack in from the range to simulation, the need for believable user activity increases. GHOSTS client software was designed to replicate a wide variety of user activities on computers and networks. An office administrator might edit documents, send email, coordinate with colleagues, browse news sites, and download files. A hostile actor might perform reconnaissance, move laterally, stage data, or use communication channels to coordinate. Hostile actions are more exciting, but mundane activity is what makes the range feel real. We include both to provide a realistic, challenging environment for defenders to use as they practice locating threats in the hum of the everyday.

As our exercise work matured, the idea of mission rehearsal became more important. In a mission rehearsal, the simulated environment should give teams a chance to practice the operational problem they may face in the field. For traditional defensive cyber operations, that may mean defending hosts, detecting malware, hunting across telemetry, containing compromise, and recovering systems. For OIE, that may also mean understanding how a technical disruption affects public perception, how adversary messaging exploits uncertainty, and how counter-messaging or recovery actions alter the trajectory of the scenario.

This is in keeping with the original GHOSTS idea, just at a larger scale. The wargaming literature identifies the same gap from the exercise side—the information environment is often underdeveloped or underrepresented in games, and adjudication methods still struggle to capture cognitive, human, and non-

attritional effects [Paul 2020, Wilden 2023]. As the information environment has gotten more complex, so has our sense of what sufficient reality looks like.

2.2 Separating the Agent from the NPC

One of the earliest architectural distinctions in GHOSTS was the separation between the agent and the NPC. The agent was the binary running on the host, driving the creation of activity. The NPC was the identity model that determined which activity was created, and included parameters like role, activities, and scenario function. Over time, the NPC grew to encompass relationships, knowledge, preferences, beliefs, motivations, and social context.

This distinction let us separate execution from the content of what was executed. GHOSTS could drive a machine to perform actions, but the activity was driven by the details of each NPC. That distinction became more important as the project matured. At first, the NPC might only need a role and a schedule. Later, the NPC needed much more complexity. The agent remained the execution mechanism, and the NPC became the model of why those actions made sense.

The agent/NPC distinction also became essential once GHOSTS moved beyond enterprise endpoints. In blue space, an NPC may still be tied to a host, account, browser, email client, document, file share, process, or application. In gray and red space, an actor may post, communicate, influence, observe, coordinate, react, or amplify without being anchored to any known machine.⁸

This distinction also helps keep GHOSTS conceptually clean as AI enters the platform. Workflow and AI services may assist with planning, action selection, content generation, or orchestration, but the system still needs a model of the actor whose behavior is being expressed. Without it, AI-generated activity becomes just another source of noise.

2.3 Activity Generation to Behavioral Fidelity

Early success also exposed several areas for improvement. A recurring theme was the frequency and distribution of observed behavior. What looked acceptable for a

⁸ Blue space comprises friendly networks, systems, and users under the defender's control; red space comprises adversary-controlled infrastructure and activity; and gray space comprises external, shared, or third-party environments that neither side fully controls, such as the public internet, partner networks, and commercial services.

single NPC could create big-picture problems across a range. For instance, traffic could become too predictable, or the breadth of sites browsed could be too narrow.

Some randomness looked synthetic precisely because it was random in the wrong way. Real users may differ from each other, but their individual actions are not random. They develop habits, return to familiar websites, follow recurring interests, linger on some pages longer than others, and move through the web along recognizable paths. A user who visits a random assortment of sites may generate traffic, but not a convincing pattern.

We needed NPCs with enough variation to avoid repetition, but enough continuity to reflect the habits and interests of a plausible person. To support this, we built up our NPC definition, adding preference structure, better browsing logic, website categorization, and stickiness. This work became SPECTRE [Updyke 2021], which considered the patterns of real people who have preferences, habits, routines, and context, but whose actions are not mechanically repetitive.⁹

Earlier agents would start at the top level of a site, browse down several levels, and then move to another site, which generated traffic but did not look like modern web behavior. Real people arrive at deep links from search engines, email, instant messages, documents, social media, and previous browsing history. They access long URLs, return to familiar sites, skim, pause, download, abandon pages, and follow interests over time.¹⁰

The same lesson applied to payloads. If only a few documents are downloaded or uploaded during an exercise, each becomes suspicious and exfiltration-like activity becomes too easy to identify. Real networks contain constant mundane movement of PDFs, Word documents, spreadsheets, slide decks, archives, images, videos, installers, and other artifacts. By including that baseline of activity, we make an environment that is more realistic and more of a challenge for defenders.

2.4 From Synthetic Activity to Synthetic People

Around the same time, GHOSTS started shifting from generating actions to generating people. ANIMATOR pushed NPCs toward even richer synthetic

⁹ <https://github.com/cmu-sei/GHOSTS-SPECTRE>

¹⁰ An example of the unintended outcomes of realistically mirroring human behavior is by launching many tabs to read later, NPCs would create resource problems on resource-constrained ranges. This prompted many GitHub discussions on behaviors that perhaps we should not attempt to replicate.

identities, including biographical detail, relationships, job history, financial attributes, organizational context, and insider-threat characteristics ¹¹.

Once personas became richer, user activity became the behavioral surface of a synthetic identity. A browsing session, login event, document edit, message, file download, or social media post could be interpreted as something a particular person did in a particular context.

This was a significant shift in the project, and with it, it became easier to later reason about social and cognitive realism. A person has preferences and relationships, knows some things but not others, and may have motivations that make some actions more likely than others.

The work on GHOSTS to create preferences allowed agents to browse based on individual interests rather than generic patterns. A preference is a small claim about what an actor is more likely to do (or not do). If an agent liked a sport, team, topic, or hobby, that preference could alter their site selection and browsing behavior. This was a simple but powerful change that created a basis for decision making.

Later additions made the model more expressive. Motivations help explain why an NPC prioritizes certain actions. Relationships help model advice seeking, trust, information sharing, social influence, and interpersonal dynamics. Knowledge helps determine what an NPC can discuss, search for, misunderstand, or notice. Beliefs help model interpretation, disagreement, persuasion, polarization, and reaction to influence. Together, these attributes enable an NPC to behave as an actor inside of a world.

2.5 Expanding the Enterprise Surface

GHOSTS gained many new capabilities in the version 6/7 era, including Jabber/XMPP, RDP, sFTP, SSH, AutoIT, cron-like scheduling, new mail functions, and additional options for scripted browsing patterns. We also expanded administrative control so that range administrators could start and stop timelines, use timelines as default activity patterns, and introduce just-in-time activities. A great deal of browser-based development also happened during this period, including Linux browser support, crawl mode, and headless browsing improvements.

¹¹ <https://github.com/cmu-sei/GHOSTS-ANIMATOR>

The pattern described in Table 1 repeated many times when developing GHOSTS—solving one realism problem revealed the next one. As one part of the environment started looking better, something else looked thin. GHOSTS continued advancing because range realism kept exposing the next missing layer.

Table 4: *Realism Boundary Expansion*

Realism Problem	GHOSTS Response	New Problem Exposed
Cyber ranges felt empty	Add synthetic user activity	Activity looked mechanical
Activity looked mechanical	Add schedules, variation, and realistic application use	Randomness looked synthetic
Randomness looked synthetic	Add preferences, stickiness, and behavior patterns	Users lacked identity
Users lacked identity	Add persona models	Personas needed relationships and context
Personas needed relationships and context	Add social structure, motivations, knowledge, beliefs, and trust	Actors needed a world to inhabit
Actors needed a world to inhabit	Add Pandora/content environments	Content needed interaction and propagation
Content needed interaction and propagation	Add posts, sharing, offline interactions, social dynamics	Manual scripting became too brittle
Manual scripting became too brittle	Add LLM-assisted action selection	Model output needed control
Model output needed control	Add workflows, deterministic handlers, telemetry, and operator constraints	Mission rehearsal needed cross-layer traceability
Cross-layer traceability was needed	Expand observability across cyber, social, and cognitive layers	Future work: Make effects measurable, reproducible, and explainable

2.6 Building Content Environments with Pandora

Pandora¹² emerged from a practical realism problem. GHOSTS could make users act, but they lacked anything to interact with, especially in closed or air-gapped environments. Pandora filled this gap by serving the informational terrain around the NPCs, including websites, documents, PDFs, images, media, archives, and other objects, with support for controlled red-team payloads when needed. In that sense,

¹² <https://github.com/cmu-sei/GHOSTS/tree/master/src/Ghosts.Pandora>

Pandora was part of the move from simulating user activity to simulating a usable world.

In an air-gapped environment, URL accessibility is a deeper problem than it first appears. On the live internet, a resource must exist behind a URL, or the user receives an error. Inside a closed range, agents need URLs that resolve to plausible content, which meant that users needed to maintain long lists of valid URLs by hand—a brittle and tedious process. Pandora reduced that burden by responding to requests with appropriate generated or hosted content, including document repository-like behavior like OwnCloud, Box, or SharePoint. This meant that agents could browse random URLs and download diverse payloads without every object being manually staged in advance.

Pandora also helped with payload realism. By configuring clients to download assets from a Pandora server, exercise designers could increase the number and variety of downloads across the range. That made malicious or suspicious downloads harder to spot and gave defenders a more realistic baseline to analyze.

But that improvement revealed another edge—uploads. If agents downloaded frequently but rarely uploaded, exfiltration or outbound transfer still stood out. Real users upload documents, images, links, forms, attachments, archives, and snippets of text throughout the day. Social media made this issue more obvious. People do not only consume content; they create and share it.

Table 5: Evolution of Pandora from Content Server to Information Environment

Stage	Realism Gap	Pandora Evolution	Key Capabilities
1. Usable Content	NPCs could act, but closed ranges lacked a believable world for them to interact with.	Generated and hosted content	Websites, documents, PDFs, images, media, archives, and other downloadable objects
2. Air-Gapped Browsing	Manually maintaining valid URLs inside closed environments was brittle and labor-intensive.	Dynamic content resolution	Plausible content returned for requested URLs; repository-like behavior modeled after OwnCloud, Box, or SharePoint
3. Payload Realism	Suspicious downloads stood out when ordinary users rarely downloaded diverse files.	Background download activity	Higher volume and variety of benign, suspicious, and controlled red-team payloads

Stage	Realism Gap	Pandora Evolution	Key Capabilities
4. Participatory Activity	Download-heavy users still looked synthetic because real users also upload and share content.	Uploads and content creation	Documents, images, links, forms, attachments, archives, and text snippets
5. Information Environment	Social and influence scenarios required content to move among simulated actors.	Shared and measurable informational terrain	Posts, media sharing, NPC-to-NPC visibility, scenario labels, coordinated behaviors, and measurement targets

Consequently, Pandora’s role expanded from serving content to creating a broader content environment. Social media-like services could accept posts, links, documents, and media from agents. One NPC could post something another NPC could see.

The information-disorder and influence-operation literatures help name this terrain: content can misinform, manipulate source credibility, exploit social proof, simulate consensus, or move through coordinated behaviors [Wardle and Derakhshan 2017, DISARM Foundation 2023]. Longtchi and colleagues make the same idea more measurable for malicious emails by distinguishing psychological techniques and tactics [Longtchi 2024]. For GHOSTS, these taxonomies are useful as scenario controls, content labels, and later measurement targets.

2.7 Content to Social Interaction and Agent Reasoning

Once the content environment existed, a new question appeared—what should agents post, share, discuss, or react to? A social post expresses a person’s context, interests, beliefs, relationships, mood, knowledge, and goals, and it may inform, persuade, complain, coordinate, joke, mislead, amplify, panic, or reassure.

Early if-then logic could cover narrow cases. If it is an agent’s birthday, the agent might post about dinner with friends. If the agent likes a team, the agent might post about a game. But this approach becomes tedious quickly as it requires exercise designers to anticipate too many situations, configure too many templates, and accept limited variability.

The richer agent reasoning framework addressed this by giving NPCs more internal structure. Preferences allowed agents to favor topics, sources, sites, and content. Motivations helped explain why they might prioritize status, curiosity, order, social contact, idealism, independence, or other drives. Relationships allowed information to move through trusted or familiar ties. Knowledge gave agents something to know,

learn, share, or misunderstand. Beliefs allowed agents to interpret information differently, defend positions, disagree, or become susceptible to influence around specific topics.

This is also where offline experience became important. Much online activity is shaped by offline interaction like meetings, hallway conversations, water cooler exchanges, classroom discussions, workplace rumors, and casual encounters. Initially, agents lacked this background. To provide a more realistic foundation for decision making, we created an offline job server to simulate interactions where agents might share information on specific topics. These interactions could change knowledge, beliefs, preferences, and later actions.

ANIMATOR became the natural place for this capability. It could create agents at the beginning of an exercise, but it could also continue to enrich their profiles during the exercise through agent animations. Operating on ticks or cycles (much like modern agent-based modeling systems), ANIMATOR could evaluate the current state of agents and their environment, determine whether interactions should occur, log outcomes, and update profiles. If two agents had a strong relationship and shared interests, the system might simulate a friendly exchange. If two agents had opposing beliefs, the interaction might become a debate. These events could influence later browsing, messaging, posting, and interpretation.

The architecture for social posting preserved the GHOSTS pattern. ANIMATOR evaluated agent state and generated social activity based on preferences, motivations, relationships, knowledge, and beliefs. The GHOSTS API received the activity and queued it for the appropriate client. The client executed the action on the designated platform and reported back. This allowed richer behavior to be added without requiring every existing installation to abandon its core infrastructure.

This line of work is important for cyber exercises, but it becomes especially important for rehearsal of OIE. Influence operations depend on exposure, interpretation, trust, repetition, social reinforcement, and behavior change. Cognitive-warfare scholarship underscores that the target is decision advantage, and useful measures include decision latency, decision error, recovery time, and trust calibration [du Cluzel 2020; Rushing 2024; Rushing 2026; Rushing and Xu 2026]. If NPCs have no beliefs, no trust structure, no relationships, and no memory, influence becomes a scripted inject rather than a simulated effect. A richer NPC model lets the exercise represent how information moves, how actors interpret it, and how social and cognitive state changes over time.

2.8 Using LLMs for Action Selection

Around this same period in 2023, we found another opportunity to improve the fidelity of the system by using large language models [Updyke 2023]. The question was whether LLMs could help decide what an NPC should do next and why. We evaluated if GPT-3.5 Turbo and GPT-4 could generate activity directives for GHOSTS NPCs, and the result was clear—LLM-generated activity was more complex, more diverse, and more specific to the persona than the data created by earlier native logic.

This also pointed at a long-standing pain point in implementation: building complex timelines by hand is tedious, brittle, and hard to scale. It requires scenario designers to anticipate what an NPC should do, how that behavior should vary across time, and how it should change based on context, role, personality, relationship, belief, and scenario state. LLMs offered a different path. Instead of manually scripting every action, we could describe a synthetic person, including their situation, their goals, and their constraints, then use the model to help generate plausible or even likely next actions with reasons attached.

This work showed that LLM directives go beyond content generation and could become a control surface for action selection. If Pandora helped create the information environment around synthetic users, LLM directives pointed toward a future where those users could reason more plausibly about that environment. This matters for OIE mission rehearsal because adversaries act through language-rich channels. Rumors, public statements, spoofed alerts, social-media posts, news narratives, chat messages, emails, and counter-messaging all require content that has context. LLMs can help generate this content and can help NPCs react to it in ways that reflect their attributes. But the model output must still be constrained by scenario logic, range administrator control, observability, and repeatability.

2.9 Model Output to Workflow Control

At the same time, the LLM work exposed operational problems like cost management, output variability, translation difficulty between natural language and executable directives, and skewed outputs. The answer has been to treat LLM control as part of a broader workflow substrate rather than as an unconstrained replacement for prior coded logic. Even with LLMs providing variation and depth in activity creation, GHOSTS still needs deterministic execution, observable state, and operator control.

Workflow control is especially important when simulation works across cyber, social, and cognitive layers, creating complex interactions. For instance, a cyber event may change what an NPC can access, a social event may change what an NPC has heard, or a rumor may amplify through certain relationships but fail in others. These dynamics cannot be handled well by a flat activity timeline.

The goal was never to make every NPC fully autonomous. The goal is to make their behavior stateful, interpretable, and controllable. Workflows can coordinate the parts, like scenario conditions, actor models, LLM calls, deterministic handlers, telemetry, and range administrator decisions. This is close to Xu's cybersecurity-dynamics framing, where security is treated as a time-dependent interaction among attackers, defenders, users, systems, uncertainty, and emergent population behavior [Xu 2014, 2019, 2020]. The platform can then preserve the values that made GHOSTS useful in the first place—real execution, observable evidence, repeatability, and operator control.

2.10 Consolidating the Platform

As seen in Table 3, Version 8 consolidated a great deal of functionality. GHOSTS had originally been modular, partly because it was not obvious that every user would want or need every capability. Over time, feedback pushed in the other direction. Users wanted the default system to come with all the pieces needed to create a realistic environment. So, version 8 folded adjacent capabilities like ANIMATOR and SPECTRE into the core platform, consolidating technology dependencies, and making the framework feel less like a set of associated tools and more like an operating system for simulated human activity. The capabilities could remain configurable, but the platform needed to be coherent out of the box.

The consolidation also made sense conceptually. Preferences, personas, browsing behavior, content environments, workflows, social activity, and cognitive attributes are not separate realism projects, but layers of the same problem. When users need sufficient realism, they expect these layers to work together. The platform can remain extensible, but the default experience should communicate the whole idea.

That consolidation becomes more valuable as GHOSTS supports mission rehearsal. A scenario designer should not have to assemble a dozen tools before modeling a coordinated cyberattack with public reaction and adversary messaging. The system should provide enough integrated capability to make the first scenario possible, then enough modularity to tune it for the mission.

2.11 Cyber, Social, and Cognitive Simulation

The current work extends that same trajectory into cyber, social, and cognitive simulation. The question is no longer only what agents do, but why they do it. GHOSTS began as “General HOSTS,” but has evolved into Generalized Human-Organizational Simulation of Teammates and Systems: a platform for modeling how human and machine actors behave inside technical and organizational environments. It is a synthetic user platform with multiple client modes, dozens of activity handlers, integrated persona generation, social and cognitive modeling, and workflow-based control for producing sufficient realism at scenario or world scale.

Version 9 frames the future of GHOSTS around three interacting domains: cyber, social, and cognitive. Cognitive activity is no longer tied to machine-bound clients. In the original model, a GHOST client acted through a host, like a workstation, browser, email, document, process, or other observable endpoint. This use case remains important for blue-space users operating inside an enterprise or range environment.

Current work expands the model to include actors whose behavior may not be anchored to a known machine. Gray and red space users may post, communicate, influence, observe, coordinate, or react without us knowing exactly which host they are using or where that activity originates. Scenarios and executions become the organizing layer across cyber, social, and cognitive activity. GHOSTS is no longer only simulating endpoint behavior; it is increasingly modeling the relationships, beliefs, knowledge, preferences, trust, and influence processes that explain why actors behave the way they do, regardless of whether their activity begins with a keyboard and mouse.

The previously discussed multi-stage cyber-social-cognitive mission illustrates the point. The value of this scenario is not any single layer, but the interaction and the result produced. Blue teams isolate compromises and reroute traffic. Counter-messaging attempts to contain misinformation. Infrastructure recovery competes with public anxiety. NPC sentiment and trust shift as people encounter outages, rumors, official statements, and peer reactions. The debrief reconstructs attack vectors, public reaction timelines, points where trust eroded, and which defensive actions changed the outcome.

2.12 GHOSTS and AI

The AI space has changed quickly, and GHOSTS has had to change with it. Earlier, we built GHOSTS Shadows as a server-side way to manage LLM integration and connect model outputs to GHOSTS activity. That was useful, but it reflected an earlier stage of the problem—providing GHOSTS agents with access to locally hosted LLMs, particularly inside closed or air-gapped environments. RangerAI is becoming the AI infrastructure stack around our cyber range and simulation work, the place where planning, model integration, orchestration, tool use, and exercise-level AI control can live.¹³ In that model, GHOSTS does not need to own the entire AI layer. GHOSTS remains the realism and actuation engine, the system that makes synthetic users, teammates, organizations, and public actors behave inside the simulated world. RangerAI can handle more of the AI infrastructure, which continues to evolve quite rapidly. As that ecosystem matures, GHOSTS remains focused on its core mission.

AI also changes what adversaries and defenders can do inside the simulated world. It can increase the scale and variation of content generation, adapt actions based on scenario state, support richer NPC behavior, accelerate scenario construction, and help operators explore alternate paths. The power of AI is most useful when connected to a disciplined simulation architecture including actor models, scenario constraints, workflow logic, telemetry, repeatability, and after-action review.

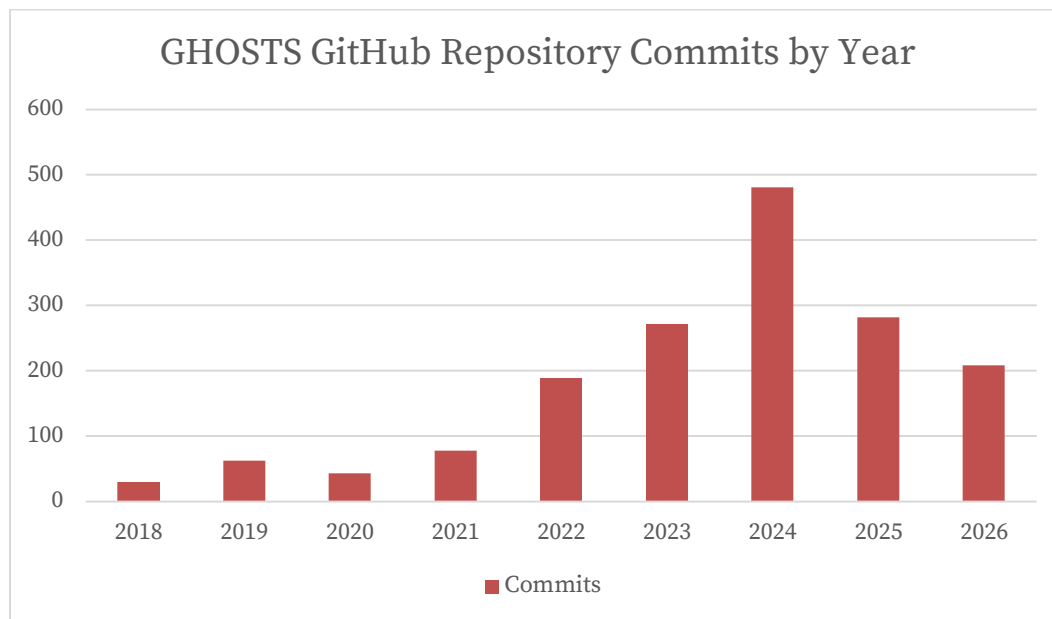
For GHOSTS, the main AI question is not whether a model can generate a plausible sentence or activity directive. Instead, we are concerned with how a generated action fits into a simulation. What state did the model see? Why did the NPC act? What changed afterward? Was the action executed? Was it observed? Did it affect another NPC, a defender, a public narrative, or a mission outcome? By focusing on these impacts, we work to use AI output in a methodical, focused manner.

¹³ <https://github.com/cmu-sei/rangerai>

3 Lessons from Open Source Sustainment (OSS)

We have learned many lessons from GHOSTS being OSS. The first is that the repository becomes part of the system. The code matters, but so do installation paths, examples, release notes, issues, discussions, contribution rules, and documentation. It's important that an outside user can understand what the project is and where it's going.

GHOSTS has matured in the public sphere through releases, documentation, issues, discussions, and feature requests. Those artifacts are easy to treat as administrative overhead, but for us they became part of the research and evidence trail. They show how the project has changed, what claims we made about it, what users struggled with, and where the architecture created friction.



Peak: 2024 — 481 commits, ~2.5× the project average. Coincides with the AI build-out.

Tempo: Avg. 183 commits/year. Activity has roughly doubled every two years since 2020.

Coverage: Near-continuous engagement, with less activity in the very early years.

Figure 3: GHOSTS GitHub Repository Commits by Year

This lesson becomes more important as GHOSTS becomes broader. A narrow tool can tolerate narrow documentation. A platform for cyber, social, and cognitive simulation cannot. Users need to understand the concepts, architecture, defaults, data flows, deployment assumptions, and limits. They need good examples that work and explain. They need enough confidence to try the software in their own contexts.

3.1 Building in Public

OSS forced a different kind of honesty from our team. Internally, a research prototype can survive on tacit knowledge, hallway explanations, and the fact that the people running it already know its rough edges. A public project does not get that luxury.

If the documentation drifts from the code, users feel it immediately. If releases, examples, and configuration files are out of sync, it may render the project inaccessible to those that would otherwise be interested. If the architecture is too modular, new users may struggle to assemble a working system. If everything is bundled together, maintainers inherit a larger surface area. If any or all of these items are left unchecked, it can result in an untenable support burden. GHOSTS has now lived inside that tradeoff for years.

We remain committed to the spirit of the OSS community and to supporting use cases we could not have envisioned when the project began. That is one of the surprising outcomes of making GHOSTS public. Users find use cases we did not anticipate. Some need agents to do highly specific things at highly specific times, which pushes against the project's early bias toward randomness. Others need lighter installs, different operating systems, different activity handlers, or different deployment assumptions. Every one of those requests becomes a test of the system's capability.

OSS exposed a tension between research ambition and user experience.

Researchers often want to preserve flexibility because the next experiment is unknown. Often, users want a clear path because they are trying to solve today's problem. GHOSTS had to mature through that tension. It needed enough flexibility to keep supporting research and enough coherence to be useful as software.

Moreover, a public project must show its assumptions. What does the NPC do? What is modeled? What is random? What is deterministic? What can be observed? What must be configured? What is still missing? OSS makes those questions impossible to avoid.

3.2 Sustainment Is Engineering Work

The practical lesson is that OSS sustainability is work. It is versioning, documentation, installation, examples, CI, release notes, support, and dependency management. Most of all, it is patience. Many OSS projects do not take off

immediately. They gain users because maintainers build a history of responsiveness. Questions get answered, bugs get fixed, issues get triaged, and users can see that the project is alive. We believe that responsiveness matters. When people evaluate an OSS project, they are not only evaluating code. They are evaluating whether the project looks active, whether the maintainers appear reachable, and whether the community has somewhere to go. A repository with no activity for six months raises key questions before anyone bothers to look at the architecture.

Maintenance also requires emotional distance from the work. Users will complain. They will find bugs. They will ask for things the system does not do. The maintainer must separate criticism of the project from criticism of the person, because every issue surfaced is an opportunity to improve the software and build trust with the user community.¹⁴

For GHOSTS, being OSS has been more than a distribution model. It has been a forcing function. It made the platform more legible, exposed integration pain, created a feedback loop with real users, and turned maintenance discipline into part of the mission.

Continued development has also enabled realism to compound without collapsing under its own weight. Every new capability creates new documentation, support, dependencies, defaults, edge cases, and user expectations. If sustainment and patience do not keep pace, the platform becomes impressive but unusable. The more ambitious GHOSTS becomes, the more ordinary engineering work matters.

3.3 Simplification Requires Taste

Simplification is hard. It requires knowing the domain, the users, the software, and the likely direction of the system. It is easy to add options. It is harder to decide which options should become defaults, which should become configuration, and which should be removed.

Someone once told us that every project needs at least one engineer with good taste. That has proven true. Good defaults are not just technical choices. They encode

¹⁴ OSS work is very much in the spirit of one of our favorite quotes about cyber more generally: *“Cyberdefense isn’t magic. It’s plumbing and wiring and pothole repair. It’s dull, hard, and endless. The work is more maintenance crew than Navy SEAL Team 6. It’s best suited for people who have a burning desire to keep people safe without any real need for glory beyond the joy of solving the next puzzle.”* [Wheeler, 2018] We do not, however, find it dull.

judgment about how most users should experience the system.¹⁵ You cannot support everything everyone wants, but the work is to know the domain well enough to make decisions that serve most users without closing off serious use cases.

This is why GHOSTS moved away from some forms of modularity and toward a more integrated install. The litmus test became simple: can a user pull this down, run a command, and see the software work? If the answer is yes, the project becomes easier to evaluate, easier to trust, and easier to extend.

Simplification also protects the conceptual model. A project like GHOSTS can easily become a catalog of capabilities, options, subprojects, and configuration files. That catalog may be technically impressive, but it can obscure the main idea. The user should be able to understand what the system is for, which is creating sufficient realism through believable synthetic activity, modeled actors, and observable execution.

AI features can create the impression that anything is possible, but useful simulation still requires constraints. Good taste means choosing where AI belongs, where deterministic code belongs, where configuration belongs, and where the operator must remain in control. The easier answer is to add another option, but the better answer is to make the system coherent.

3.4 Observability Is Part of Realism

GHOSTS was never only about making activity happen. It was about making activity visible, controllable, and explainable enough for range administrators and participants to use. The API, web interface, timeline model, and reporting mechanisms mattered because they allowed synthetic activity to be coordinated and inspected.

This becomes more important as GHOSTS moves into cyber, social, and cognitive simulation. If an NPC downloads a document, posts a rumor, changes its belief state, reacts to a message, amplifies a narrative, or loses trust in an institution, the system needs to preserve enough telemetry to reconstruct what happened. Otherwise, the exercise becomes theatrical rather than analytical.

¹⁵ A good analogy of setting smart defaults is outlined by Hansson, “Rails is omakase. A team of chefs picked out the ingredients, designed the APIs, and arranged the order of consumption on your behalf according to their idea of what would make for a tasty full-stack framework.” [Hansson 2012]

Mission rehearsal requires after-action review. After-action review requires traceability. Traceability requires instrumentation. Operators need to know which agents acted, which timelines or workflows were active, which content was seen, which posts were generated, which messages spread, which beliefs or trust values changed, which services failed, and which defensive actions altered the scenario.

Observability also supports credibility. Sophisticated participants will ask why something happened. Exercise designers will ask whether a model behaved as intended. Researchers will ask whether a result can be reproduced. Sponsors will ask what was learned. GHOSTS attempts to address these questions through time-aligned telemetry across the environment, enabling conclusions to be grounded in data rather than anecdote.¹⁶

The next stage of GHOSTS therefore needs telemetry that spans technical, social, and cognitive layers. Packets and logs remain important, but so do exposure histories, engagement histories, sentiment shifts, belief changes, relationship-mediated spread, and reconstructed mission timelines. Decision-centric measures from the cognitive warfare literature can sit beside training-transfer measures from the learning literature, encompassing decision latency, decision error, recovery time, trust calibration, confidence shift, pre/post task performance, delayed retention, inter-run variance, seed sensitivity, and expert-rater agreement [Rushing et al. 2026, Salas et al. 2012]. Our goal will be to make agent behavior useful evidence in determining what has happened.

3.5 Longevity Requires Shared Ownership

The project should outlive any single maintainer. Anything that is designed and engineered with care should be able to survive beyond the person who started it. If the ideas are sound and the approach is useful, others should be able to feel ownership over the project and carry it forward.

Projects are more than their code. They require documentation, architecture discipline, contribution paths, release hygiene, and a community that understands the system well enough to improve it. They also require accepting that successful projects will eventually be used in ways the original team did not imagine.

¹⁶ While automation of human-machine team performance remains difficult, we believe continued work in self-assessment will prove promising. (Updyke 2024)

Small changes compound over years. Maintenance is typically undervalued relative to novelty, but maintenance is how infrastructure succeeds. In many cases, taking an old idea and engineering it well has more impact than inventing a new one.

GHOSTS has survived because it kept improving, responding, and finding the next edge of simulation.

Shared ownership also matters because the realism boundary will keep expanding. A platform this broad can't be built and maintained by the same small team forever. It requires a community of contributors who can each improve part of the system while preserving the overall direction, particularly in the age of AI-enabled software development. Our goal has been to ensure that the architecture, documentation, examples, defaults, and community make it possible for others to extend the platform responsibly.

4 Summary and Future Direction

Making GHOSTS more realistic meant that, with every iteration, we needed to represent a larger slice of reality. We started by creating timelines, then preferences, then personas which needed content, social structure, beliefs, knowledge, workflows, and cognitive effects. But solving these problems led to new, subtler issues, which meant we needed to add background users with plausible activities, preferences, and identities including relationships, context, and memory.

We set out to create NPCs, but we came to understand that NPCs on their own cannot create a sufficiently realistic platform. We also needed to represent some part of the complex world they inhabit.

Sufficient realism started as a practical engineering goal: to make the range feel inhabited enough to support training and experimentation. But over time, sufficient realism became a systems problem. It required endpoints, applications, traffic, documents, websites, organizations, social ties, beliefs, motivations, trust, workflows, and observable consequences that cohered well enough for sophisticated participants to stay immersed.

People have always asked us a simple question: *How do you know what is realistic?* After a decade of GHOSTS, our answer is different than it would have been at the beginning. Realism is not something we define once in the architecture. It is something we discover through use. We know a behavior is insufficient when participants notice it, when defenders can game it, when telemetry looks wrong, when reports show traffic is too predictable, when POST activity is too thin, when browsing is too narrow, when timelines spike unnaturally, or when the environment fails to support the training objective.

At the same time, GHOSTS cannot and does not need to simulate everything. It needs to simulate enough of the right things, at the right fidelity, for the purpose at hand. Our aim is not for a fully operational simulation of every possible scenario, but *sufficient realism*. Cyber hunt exercises, insider threat scenarios, and malware analysis ranges all have different realism needs. The requirements are not fixed. They depend on what participants are supposed to believe, observe, decide, and do.

Over the last decade, GHOSTS kept finding that our expectations about what matters in simulations were insufficient. More of the human, organizational, and informational environment mattered than we initially expected, and so we built the system to enfold more of these factors over time.

The next decade of GHOSTS will likely be shaped by OIE, because operational threats have expanded. Teams must now be able to rehearse missions in which cyber disruption, public reaction, adversary influence, organizational uncertainty, and AI-enabled behavior interact. Teams must be able to practice defending both a network and a social narrative. This requires several lines of work.

First, GHOSTS needs richer actor models. NPCs should continue to represent roles, preferences, relationships, knowledge, beliefs, motivations, trust patterns, memory, and decision logic, but these attributes should not be treated as ornamental persona fields. They should affect what actors see, how they interpret it, what they do, and how they influence one another.

Second, GHOSTS needs expansive environment models. Synthetic actors need content ecosystems, social platforms, communication channels, enterprise services, infrastructure dependencies, public-facing information sources, and believable terrain to move through. Pandora and related content services point in this direction, but OIE mission rehearsal will demand broader information ecologies.

Third, GHOSTS needs stronger orchestration. Scenarios need to coordinate cyber events, infrastructure effects, public reaction, adversary messaging, defender response, counter-messaging, AI-generated activity¹⁷, and cognitive state changes under operator control. Static timelines will remain useful, but complex mission rehearsal requires workflows that can respond to state.

Fourth, GHOSTS needs deeper telemetry. The system must reconstruct what happened across technical, social, and cognitive layers, including who saw what, who believed what, who acted, who influenced whom, what changed, where trust eroded, where defenses worked, and where the scenario tipped. Without that telemetry, the simulation may feel immersive but remain hard to learn from.

Fifth, GHOSTS needs disciplined AI integration. AI can help generate content, select actions, vary behavior, and reduce scenario authoring burden. But AI output must be constrained, observable, explainable, and connected to executable handlers and scenario state.

Sixth, GHOSTS needs continued sustainment. The platform must remain installable, understandable, extensible, observable, and maintainable as its realism boundary

¹⁷ For GHOSTS to serve as an effective tool for training and exercise, deterministic activity is likely preferred due to explainability and repeatability needs. In wargaming and OIE, AI's non-deterministic nature could greatly enrich the simulation, but what can teams learn from the results of non-deterministic behaviors? This is an ongoing tension that future work should continue to address.

expands. OSS discipline, documentation, examples, defaults, and shared ownership are central tasks that will allow this work to survive.

The next decade of GHOSTS should be judged by how well we manage that expansion—what we can model, what we can measure, what we can reproduce, what we can explain, and what we can make useful at scale.

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14. ABSTRACT (maximum 200 words) GHOSTS began in 2017 as a practical response to a persistent cyber range problem: realistic cyber exercises require networks to be filled with traffic from users and supporting infrastructure. This provides data for exercise participants to analyze but also provides a way for the adversary to blend in and hide its activity. Over the past decade, the project evolved from a host-based activity generator into a broader platform for simulating human, organizational, and machine behavior across cyber, social, and cognitive layers. The central design principle of GHOSTS is sufficient realism. The system does not attempt to reproduce human behavior perfectly. Instead, it creates believable activity at the right fidelity to support training, experimentation, assessment, and mission rehearsal. This paper traces the evolution of GHOSTS as it grew from producing enclave activities (and the resultant network traffic, logs, endpoint artifacts, and browser activity) to creating non-player characters (NPCs) with preferences, relationships, knowledge, motivations, beliefs, workflows, content environments, and LLM-assisted decision making. It also describes the engineering lessons learned from sustaining GHOSTS as open source research software, including modularity, dependency management, documentation, observability, defaults, maintainability, and shared ownership. As cyber operations increasingly intersect with broader operations in the information environment, GHOSTS now supports the larger mission of helping teams rehearse ideologically contested scenarios.		
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